

CLOUD WARS EXPLAINED

The market for cloud computing refuses to act according to expectations! Since years Silicon valley giants are trying to make it big in the cloud services space, but just cannot go pass Amazon. What advantages did Amazon get just to enter this space first, everything decoded and explained here: <http://dgit.in/CloudWrs>

UNSEEN PART OF START-UPS

Promising new technological ideas give birth to a new startup quite often, but only a few survive till the end seeing the light of the day. Behind the good and bright lies a dark side of startups that has little dirty secrets. Find out more. <http://dgit.in/darkstartup>

DON'T INSTALL APPS, STREAM THEM!

Google's latest experiment is to let you use the Android app by streaming it off its servers without installing it on the device. It comes on the Beta version of the Google Search app. Search for an app, and if it is indexed, you will see an option to stream it now! Go check it out. <http://dgit.in/GStreamApp>

GET YOURSELF A HOVERBOARD!

Around the globe tech companies are launching new hoverboards and this is one thing that has spread like a wildfire in a forest! Let it be for going to work or going out for a movie, hop onto one, and go around. Wondering why are hoverboards everywhere right now, hop on and start reading! <http://dgit.in/AllAbtHB>

its standard spherical magnifying lenses (we estimate) and projected into the viewer's eyes. There was talk of having double Fresnel lenses in an upcoming iteration, but more on that later.

Simulator sickness or the nausea that many 3D VR users experience has been dealt with quite cleverly within the Tesseract. While we didn't really game for hours on end, whatever little time we spent wearing the HMDs, we didn't experience any physiological side-effects that are all too well-publicized elsewhere.

How does Tesseract work its magic?

The answer lies partly in its software. According to Vrushali Prasade, Co-Founder and Software Lead, Absentia VR, "In simple terms, our software converts 2D images / visuals to immersive 3D. We read all the pixels being displayed in real-time and map it values intelligently to create the illusion of depth." The Tesseract's custom software that runs on the host PC to extract frames is capable of reading image back buffers (for a visual scene) and sending that information simultaneously to the HMD as well as the traditional display monitor.

Another intelligent way to reduce nausea is done by black frame insertion and false nose illusion, according to Absentia VR. On the hardware side, Absentia VR's Tesseract incorporates two 9-axis IMUs or Inertial Motion Units as opposed to just one that's found on most generic VR headsets, thereby eliminating visual lag.

The team's software prowess is highlighted by the fact that there's not one, two but three applications that have been developed by Absentia VR to work in conjunction with the Tesseract. Other than gaming the team believes the device has applications in



other domains as well. Industrial 3D walkthroughs, visualising CAD designs in 3D and even experiencing panoramic photos and 360-degree videos which will be getting common on YouTube and Facebook.

It's no secret that virtual reality or VR is one of the most hotly anticipated evolutions of technology making inroads into our increasingly digital lives. The question isn't when as much as it is how and just how good will the experience on offer be. Let's ask the \$350 question, where essentially Absentia VR's and Tesseract's cookie effectively crumbles.

How does the Absentia VR Tesseract compare to Oculus Rift?

If we're talking about performance on a scale of 1 to 10 we'd peg the immersion at a solid 6. The controls to adjust focal length work well enough and we were able to cus-

tomise or tweak the device to three varying head sizes. We noticed a bit of warping of the rendered image and the FOV didn't feel very wide. The algorithm that creates the illusion of depth by varying the sizes of the rendered objects could also use some tweaks to increase depth perception. The head-tracking mechanism was fairly sensitive and we'd give the performance on that front an 8. None of us in the team could detect the adjustment made for the "false nose" which probably means the device likely succeeded in fooling our brains. As for the nausea we didn't try it on for more than 10 minutes at a stretch, so we can't really comment on the long term effects, but for the duration we experienced there wasn't any noticeable discomfort. Call of Duty: Ghosts (the game that we tried it on) looked strictly OK on the 1080p version of the device and while it looked better on the 1440p model, the trouble with this device was that the end of the screen was perceivable at the edge of the periphery of vision – not desirable.

For those of us who don't really have that much time or a bent of mind that's geeky enough to get our hands dirty, perhaps the ₹12,000 (US \$180) Tesseract is a good alternative (once they up the ante of course) at least for the ability to view any kind of content (not just games) in 3D. The 1440p version of the Tesseract is priced at ₹20,000 (US \$300), and currently Absentia VR is taking preorders for a booking price of ₹4,000 (US \$60) and ₹4,500 (US \$70), respectively, for the 1080p and 1440p versions.

Having spoken to the team at Absentia VR – Shubham, Harikrishna, Vrushali, Aditya and others – the Tesseract will next be showcased at CES 2016.

(with inputs from Siddharth Parwatay)

